New multi-metal cathode boosts sodium-ion battery durability

A high-entropy layered-oxide cathode has retained 90 per cent of its initial capacity after 2300 cycles in ampere-hour-class sodium-ion pouch cells, addressing structural degradation that limits battery life. Researchers at the Eastern Institute for Advanced Study in Ningbo, with collaborators from Hong Kong Polytechnic University, South China University of Technology and the Chinese Academy of Sciences, engineered the cathode using multiple metal species. The approach regulates sodium-ion diffusion and delays O3-P3 phase transitions that strain the crystal lattice. The cell delivered 162.5Wh/kg at 0.1C, a promising candidate for practical sodium-ion batteries.

Indian Navy receives indigenous TROUT underwater ROVs

Kochi-based marine robotics startup EyeROV has delivered two TROUT remotely operated underwater vehicles (ROVs) to the Indian Navy, moving the indigenous platform from trials into naval deployment. Developed in Kochi under the iDEX DISC



Kochi-based startup EyeROV's remotely operated vehicle (Credit: EyeROV)

7 SPRINT Challenge, TROUT integrates mechanical systems, electronics, embedded software, navigation, sensing, and robotic control. It is designed for underwater inspection, surveillance, reconnaissance, detection, search and recovery, and robotic intervention. The ROV underwent trials for manoeuvrability, deep diving, operation in strong currents, navigation, endurance, and operator control. Its deployment enables remote underwater operations where poor visibility, depth and hazardous conditions challenge conventional methods.

MRAM chip cuts routing time

A CMOS-integrated MRAM Ising chip with 96,000 spins has demonstrated spin updates of 0.3–1ns while solving chip-routing and wiring-layer assignment problems. Researchers at Beihang University, together with Suzhou Inston Technology and Empyrean Technology, engineered the spintronic system using voltage-controlled magnetic anisotropy to tune spin-switching probability through electrical pulse width. Each update consumes less than 40fJ. The team demonstrated global routing and layer-assignment tasks in electronic design automation, targeting complex wiring configurations. Tested on Max-cut benchmarks, the architecture achieved 1.92×10^5 solutions per second per watt, highlighting potential for energy-efficient optimisation hardware.

Indian multirotor UAV hits 498km/h

India's Ahuti Mk II electric multirotor UAV has reportedly hit 498km/h during high-speed trials, making it the fastest multirotor UAV recorded in India. Developed by Hyderabad-based Apollyon Dynamics, the platform underwent about 10 redesigns of its airframe, propulsion, and motor configuration. Unlike conventional multirotors optimised for hovering, Ahuti Mk II is designed around high-speed interception. The company tested its systems at military ranges including Babina and Gopalpur, with Indian Army support. The development targets counter-drone missions, where higher speed can shorten response time against incoming aerial threats while improving interception capability.



Indigenous Ahuti Mk II Electric Multirotor hits record 498km/h to work as counter-drone interceptor (Credit: www.electronicsforu.com)

Analogue compute-in-memory targets adaptive Edge AI

AnalogAI is developing sub-one-watt edge AI processors that can train and adapt models locally, using Silicon Storage Technology's silicon-proven memBrain SAGE analogue compute-in-memory IP. The architecture combines on-chip training and inference, reducing data movement and dependence on cloud retraining. Its memBrain Tensor In-Memory Logic Element uses SuperFlash-based cells storing up to 8 bits per cell at nanoamp-level power, alongside integrated DACs, ADCs, and neural-network computation circuits. The technology targets humanoid robots, drones, and vehicles that need rapid responses to changing environments.

Car-mounted air purifier cleans brake dust and road pollution

A vehicle-mounted purification system developed by the Korea Institute of Machinery and Materials can remove roadside PM while capturing brake-wear particles during driving. The system combines grille-mounted electrostatic precipitation with a calliper-mounted brake-dust collector. Tests showed removal of at least 95 per cent of PM2.5 and 99 per cent of particles between PM2.5 and PM10 at 30km/h, while processing about 1800m³ of air per hour.